

Quadchain Frontier Benchmark Report

Andrew Liu

Madison Zhan

Silas Wu

Stephen Hung

June 2026

Abstract

quadchain is evaluated as proof-carrying compression for multiagent systems: budget-matched compression, evidence preservation, executable receiver tasks, and verifier safety.

quadchain frontier benchmark report

this report turns the quadchain demo into a research methodology: paired downstream scoring, budget-matched baselines, role evidence audits, executable receiver tasks, bootstrap confidence intervals, and explicit provenance for public-benchmark-style adapters.

headline

- eval cases: 20 across 4 dataset adapters.
 - benchmark rows: $2,880 = \text{cases} \times \text{methods} \times \text{budgets} \times \text{seeds}$.
 - quadchain mean task score: 0.6350 with ci95 [0.5987, 0.6782].
 - raw all-budget upper-bound task score: 0.8679; quadchain all-budget stress delta vs raw: -0.2329.
 - quadchain-native budget delta vs raw at the same budget gate: 0.0500.
 - quadchain mean token reduction: 57.85%.
 - role evidence audit: payload 144/144, final prompt 144/144, leakage delta 0.
 - executable role task success: raw 50.0%, quadchain 75.0%, workflow savings 74.22%.

method frontier

paired baseline comparisons

role evidence audit

executable receiver tasks

validity notes

- this is an offline deterministic harness, not a claim that full swe-bench, longbench, ruler, or needle-in-haystack were downloaded and run.
 - public-benchmark-style adapters are labeled in every case file so we do not overclaim.
 - exact evidence scoring is intentionally paired with receiver task success; future api runs should add solver outputs and blinded llm judges.
 - proxy rows for llmlingua-2 and selective context are placeholders for apples-to-apples adapters, not official package results.

why this is better

the old eval proved that one curated packet could preserve known facts. this methodology asks the harder question: at the same token budget, across multiple task families and role routes, does quadchain preserve enough verified evidence for downstream work while rejecting unsafe handoffs? that is the research-grade version of the claim.